

CCF Lab 4 - SIEM Deployment for Threats Detection

The lab goal is to install, configure and learn a practical use cases of SIEM & XDR solution like Wazuh by imitating the actions of intruders.

Task 1 - Wazuh deployment

Install Wazuh SIEM solution in any way (deployment scenario) that you prefer. Consider a possible lab environment:

- Wazuh server (could be as a part of Elastic stack)
- Wazuh agents (workers under monitoring, at least one node, but more is better)
- Attacker host (*not mandatory*, but much better to have)

It's up to you where to create the lab environment and its nodes (locally or in the cloud). But make sure that if you are going to simulate an any attacks/threats or exploit vulnerabilities, your **attacker host and workers are in isolated network**. Thus, it's recommended do not perform any illegal actions it the cloud. **This is in your area of responsibility, be very careful and attentive!**

Bonus: prepare a different agents: Windows and Linux based

Bonus: substitute Wazuh by other SIEM solution that has the comparable functionality. Show a PoC why it's a worthy alternative.

Task 2 - Configure the SIEM

Once Wazuh is installed, make familiar with its features and manager interface (it's not mandatory to describe in the report). Then configure agents `ossec` rules to work with a various Wazuh security processing possibilities. Rules types and their amount depends on how many different Wazuh features you are going to try in the next task.

Task 3 - Play with SIEM in action

Wazuh provides a numerous possibilities for Security Operations Center (SOC) implementation such as:

- Log Data Analysis
- Cloud Security
- Container Security
- File Integrity Monitoring
- Intrusion Detection
- Security Audit and Analytics
- Vulnerability Detection
- Policy Auditing
- Monitoring Logons
- Active Response
- Incident Response
- ...

By the way, we suggest you **the default and most popular cases** of SIEM usage that you might take:

- Active Response
- Brute-force attack simulation with attacker user blocking
- Malware detection by SIEM

By simulating attacks and security threats or exploiting vulnerabilities on agents, try out as many of the above Wazuh features in action as possible. The number of features is limited only by your capabilities, resources and time. Nevertheless, try to test at least two different Wazuh features.

Regarding how to check the operability of your SIEM system and get security incidents. In fact, it is absolutely your freedom to choose which methods and techniques to use for this. First of all, they are related to the functionality of Wazuh that you plan to test. Consequently, the choice of methods and tools for emulating security threats on agents is really wide, so here we just give a some motley sketch of examples:

- [testmynids](#)
- [hydra brute force](#)
- [Atomic Red Team tests library](#)
- [Caldera](#)
- [DVWA](#)
- [Nikto web scanner](#)
- [Detection guides on Wazuh official blog](#)
- ...

At the end, also learn Wazuh Incident Response features, generate Wazuh security reports and analyze its content.

Task 4 - Log management

Setup a log rotation for your SIEM system. Show on practice that it works. Explain how logs are rotated on Linux and how disk space is managed in relation to logs.

Bonus: show an advanced skills of Kibana & Wazuh log entities filtering. For example, prepare a scenario when you need to catch some security event in Wazuh using multiple or nested searching filters and expressions.

Bonus: correlate Wazuh security events with MITRE ATT&CK techniques.

Bonus: setup security alerts notifications for your SIEM system. Show on practice that it works.

Bonus: wrap your SIEM deployment as IaC within DevOps practices.